Batching and epochs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep state timing and denominators explicit.

Separate numeric training,ID-only schedules and frozen-error fixtures;report example visits and updates alongside epochs.

Keep one habit: Assess the independent gradients,path and final values with checked keys;do not generalize from main-case order coincidence.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Distinguish an example a **batch** an **update** and an **epoch**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Four **update**s occurred. State zero was the initial model before **training**.

Opening retrieval

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Look for: Trace repeated forward loss gradient and **update** steps

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [0, 0], errors [-2, -4], MSE 10, gradient -10 and new v = 1.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: **Prediction** 3, residual 0, loss 0, gradient [0, 0].

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish architectural choices from parameter values

Reference example, not the only acceptable wording: Teacher explanation: There are 8 weights and 2 biases, total 10.

Pictured end goal

By the end, I can explain **batch**ing and **epoch**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Training a small network

Gradients with a numerical check

Parameters and architecture

For the compact alternative schedule, use two 45-minute sessions to separate counting from numerical **training**. Keep the frozen hidden layer from Week 12. Have students move paper record cards into **batch**es before doing arithmetic.

The coincident final values for the proportional-target example require the independent counterexample; do not teach that ordering is generally irrelevant.

Outcomes

Distinguish an example a **batch** an **update** and an **epoch**

Trace parameter changes across successive **batch**es

Calculate **update** counts and aggregate unequal **batch** losses correctly

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach update timing and exact aggregation with complete fixtures

> Keep main numeric pairs,count-only IDs,frozen-error rows and pictured targets separate.

Main convention,complete traces and guided key

In Week 12 every **update** used both **training** examples. We now keep the same network h = ReLU(x), y = vh, with only v trained, and divide the data into **batch**es. In this lesson each nonempty **batch** produces exactly one **update**.

An **epoch** visits every **training** example exactly once, without replacement. These rules make the counts unambiguous.

Imagine five fictional records A through E in fixed order. With **batch size** 2, the groups are [A, B], [C, D] and [E]. Keep the smaller last **batch**.

One **epoch** has three **batch**es and three **update**s; two **epoch**s have six **update**s and ten example visits.

When the last **batch** is retained, **update**s per **epoch** equal the row count divided by **batch size** rounded up. A program that drops the final incomplete **batch** follows a different rule and omits examples in that pass.

Record the rule instead of silently changing the denominator.

Use the two fictional pairs A = (1, 2) and B = (2, 4), start v = 0 and rate 0.1. For one pair with hidden value h, squared-error gradient is 2 times (vh - target) times h.

For a **batch**, average these gradients at the same current v. Apply one **update**, then recompute for the next **batch**.

Assume finite numeric pairs, integer size and finite rate. With size 1, A gives gradient -4 and new v = 0.4. B then uses 0.4, giving **prediction** 0.8, residual -3.2 and gradient -12.8, so v becomes 1.68.

With size 2, both gradients are evaluated at v = 0; their mean is -10, producing v = 1.

Main data A(1, 2), B(2, 4); start v0, rate.1	Current v before batch	Gradient at this v	New v after batch
Size1:Afirst	0	-4	0.4

Main data A(1, 2), B(2, 4); start v0, rate.1	Current v before batch	Gradient at this v	New v after batch
Size1:Bnext	0.4	-12.8	1.68
Restart size 1:Bfirst	0	-16	1.6
Then A	1.6	-0.8	1.68
Restart size 2:AandBtogether	0	Mean(−4, −16)=−10	1

The main executable uses exactly the two supplied numeric training pairs, not the five-ID counting example. Each new batch recomputes its gradient at the latest v; within one batch all per-row gradients use the same vbefore its one update. The main A/Borders happen to share final v1.68 but have different paths.

That coincidence does not prove order never matters; the pictured example uses different targets and shows different final weights. The helper requires nonempty rows and size≥1, and assumes integer size and finite numeric pairs/rate.

Guided: B at v = 0 gives gradient -16 and v = 1.6. A then gives residual -0.4, gradient -0.8 and v = 1.68. Two **batch**es each cause one **update**. Exit: both teams visit ten examples.

Size 2 produces six **update**s; size 5 produces two. Record both budgets and all **training** settings.

Complete frozen-error aggregation

Separate frozen-model squared-error fixture	Batch size	Batch MSE	Contribution to total squared error
A/Berrors1/1	2	1	2
C/Derrors4/4	2	4	8
Eerror9	1	9	9
All five cases	5	(2+8+9)/5=3.8	19
Wrong unweighted mean of means	Not one five-case average	(1+4+9)/3=14/3	Overweights the single Ecase

Separate pictured order experiment and scope

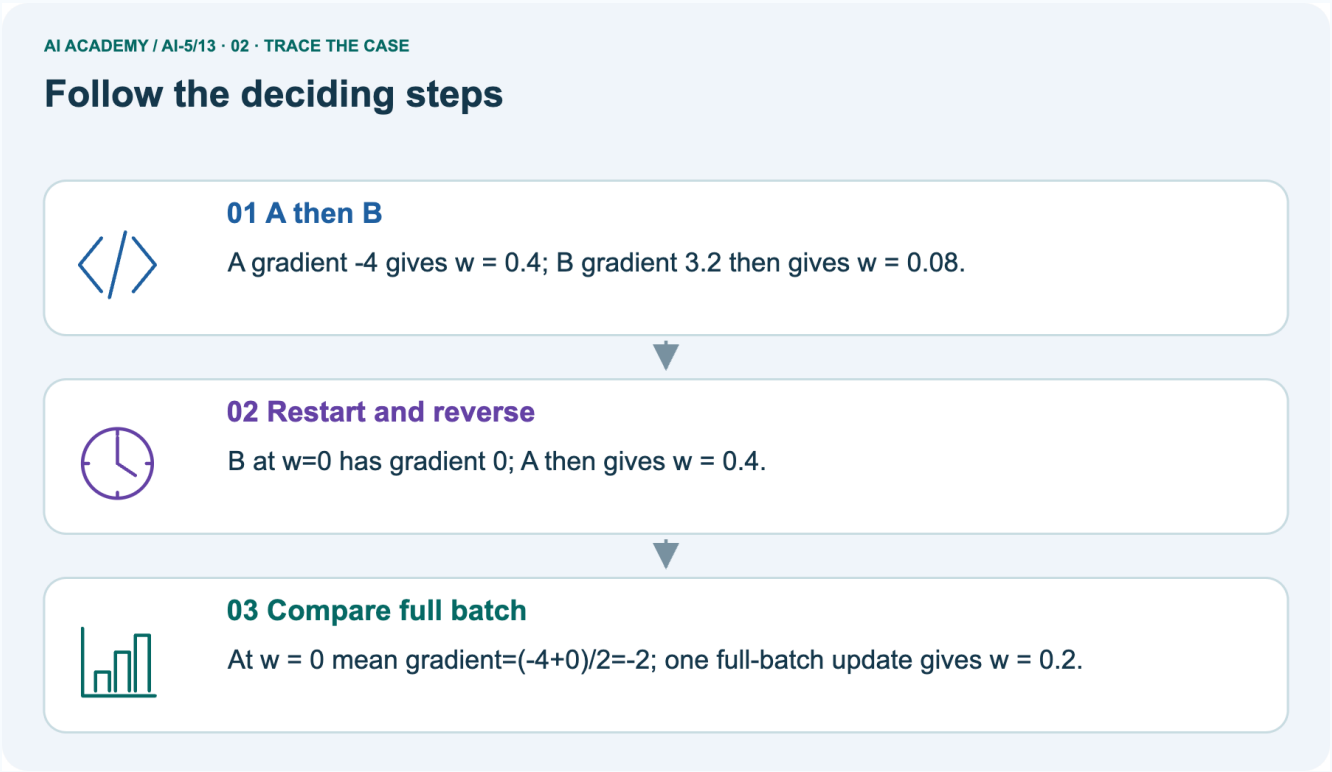
Show why an **epoch**'s example order can matter for single-example **update**s.

Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses y=wx, start w0, rate.1, A(1, 2), B(2, 0), squared loss. AthenB:size 1 gradients−4 then3.2, give w.4 then.08. Restart BthenA:gradients 0 then−4, give w0 then.4. Full-batch mean at the same old w0is(−4+0)/2=−2, giving w.2; reversing the rows leaves that exact mean unchanged. These targets differ from the main A(1, 2)/B(2, 4)and the independent A(1, 1)/B(2, 0).

Trace order dependence and full-batch invariance



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A then B	A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08.
2. Restart and reverse	B at w=0 has gradient 0; A then gives w = 0.4.
3. Compare full batch	At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

Change: Use only one full-**batch update** after reversing record order.

Prediction: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Batch	A group of training examples used together to calculate one gradient here
Update	One application of the gradient and learning rate to the current parameters
Epoch	One pass through all training examples under the stated sampling rule
Batch size	The intended number of examples in each batch with a possible smaller final batch

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
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All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

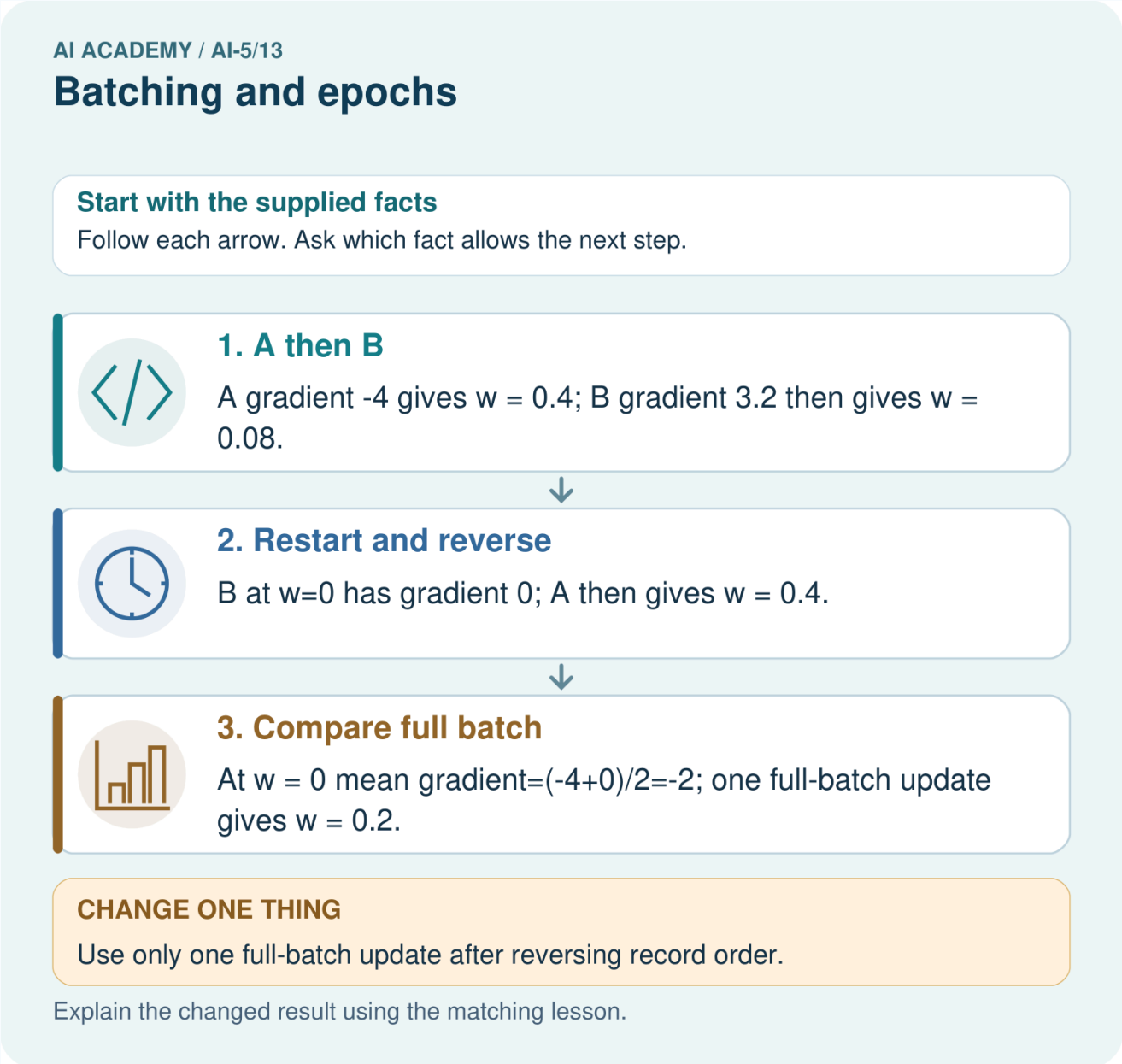
For B then A separately, show both gradients and new parameter values. Why does one epoch still contain two updates?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Batching and epochs

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Teacher check: Teacher explanation: The second gradient must use v = 0.4. Using -16 from the old state is stale; the current B gradient is -12.8 and the final v is 1.68.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: “One **epoch** always means one **update**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Only a full-data **batch** gives one **update** per **epoch** in this procedure.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The second gradient is evaluated at an already **update**d weight. Full-**batch** gradients here share the same starting weight before averaging. Changed-case check: The **update**d weight remains 0.2.

The same mean of the two starting gradients is unchanged by their order.

Changed-case reasoning: The **update**d weight remains 0.2. The same mean of the two starting gradients is unchanged by their order.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate boxes for old v, **batch** gradient and new v; allow calculators. Extension: at a fixed v, prove that the row-count-weighted mean of **batch** gradients equals the full-data mean gradient.

Explain why successive **update**s with changed v are a different procedure.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve states versus updates, then arrange five cards into batches of two.
7 to 18 minutes	Count batches, updates and visits for the table and W1; discuss retaining the final group.
18 to 32 minutes	Trace A then B at current v. Complete W2 and identify exactly where the stale gradient arose.
32 to 45 minutes	Run or trace the code and compare the full-batch update with the two singleton updates.
Session 2 start to 8 minutes	Complete the guided reverse-order trace and mark its coincidental equal endpoint.
8 to 20 minutes	Calculate fixed-model loss and use W3 to weight unequal batch means.
20 to 35 minutes	Collect W4 independently, requiring a fresh reset for the reverse-order run.
35 to 45 minutes	Discuss W5 and the exit response, then assess the fresh retry after targeted feedback.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main stale-gradient task:recompute B atv.4, gradient–12.8, final 1.68; stale–16 wouldgive2. Supported BA–16 then–.8 ends1.68. Weighted mean uses case counts; dropping 11 mod4 last3 gives8 visits/2 updates, not all rows. Independent AB:Agradient–2→v.2; Bgradient1.6→v.04. Restart BA:Bgradient0→0; A–2→.2. Both see two rows/two updates but differ in order/path/finalv. No universal strategy or future-performance claim.

Diagnose the current misconceptions

One **epoch** always means one **update**: Only a full-data **batch** gives one **update** per **epoch** in this procedure.

Reuse all gradients from the start of an **epoch**: Recompute at the parameter value before each **batch update**.

Average **batch** means equally: At one fixed model, weight each **batch** mean by the number of examples in that **batch**.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values.

Compare the final values.

Teacher explanation: A then B: A gradient -2 gives v = 0.2; B residual 0.4 gives gradient 1.6 and v = 0.04. B then A: B gradient 0 keeps v = 0; A gradient -2 gives v = 0.2. Both see the same rows but finish differently.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Count a partial batch

For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.

Teacher explanation: **Batch** lengths 4, 4, 2; three **update**s per **epoch**, nine total **update**s and 30 example visits.

W2 Repair a stale gradient

For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Teacher explanation: The second gradient must use v = 0.4. Using -16 from the old state is stale; the current B gradient is -12.8 and the final v is 1.68.

W3 Combine unequal batches

At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.

Teacher explanation: Full MSE is (3 times 2 + 10) / 4 = 4. The unweighted average 6 gives the single-row **batch** too much weight.

W5 Audit an epoch claim

A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

Teacher explanation: It visits eight rows in two **batch**es and performs two **update**s. Three rows are omitted; it did not complete a full pass under this lesson definition. Report the drop rule and actual visits.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: B at v = 0 gives gradient -16 and v = 1.6. A then gives residual -0.4, gradient -0.8 and v = 1.68. Two **batch**es each cause one **update**. Exit: both teams visit ten examples.

Size 2 produces six **update**s; size 5 produces two. Record both budgets and all **training** settings.

W1 Count a partial batch

W2 Repair a stale gradient

W3 Combine unequal batches

W4 Test order on a new case independently

Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

Teacher explanation: A then B: A gradient -2 gives v = 0.2; B residual 0.4 gives gradient 1.6 and v = 0.04. B then A: B gradient 0 keeps v = 0; A gradient -2 gives v = 0.2. Both see the same rows but finish differently.
</details>

W5 Audit an epoch claim

Independent assessment and fresh retry

W4 requires the two traces and final values 0.04 and 0.2. Fresh retry: A = (1, 2), B = (1, 0), start v = 0, rate 0.25, singleton **batch**es. A then B gives gradients -4 and 2, values 1 and 0.5.

Reverse order gives gradients 0 and -4, values 0 and 1. Ask students to name the reset and explain the order effect.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **batch**ing and **epoch**s **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish an example a **batch** an **update** and an **epoch**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate boxes for old v, **batch** gradient and new v; allow calculators. Extension: at a fixed v, prove that the row-count-weighted mean of **batch** gradients equals the full-data mean gradient.

Explain why successive **update**s with changed v are a different procedure.

Week 14 compares learning rates under one fixed **batch** rule.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/14, ai-5/16, ai-5/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Distinguish an example a **batch** an **update** and an **epoch**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Reasoned practice answer: Teacher explanation: The second gradient must use v = 0.4. Using -16 from the old state is stale; the current B gradient is -12.8 and the final v is 1.68.

Check every learner before discussion: Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: “One **epoch** always means one **update**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Only a full-data **batch** gives one **update** per **epoch** in this procedure.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use only one full-**batch update** after reversing record order. Require a **prediction** and a reason before comparing. Expected result: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

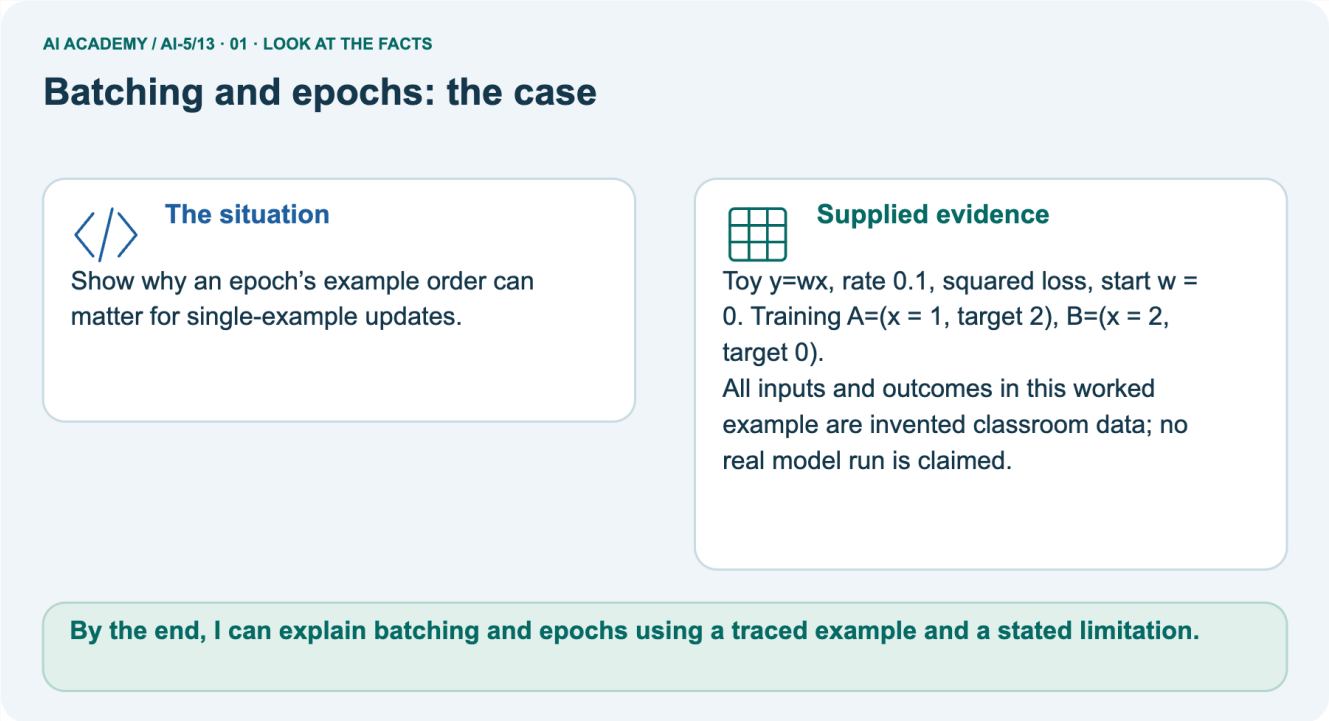
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Batching and epochs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

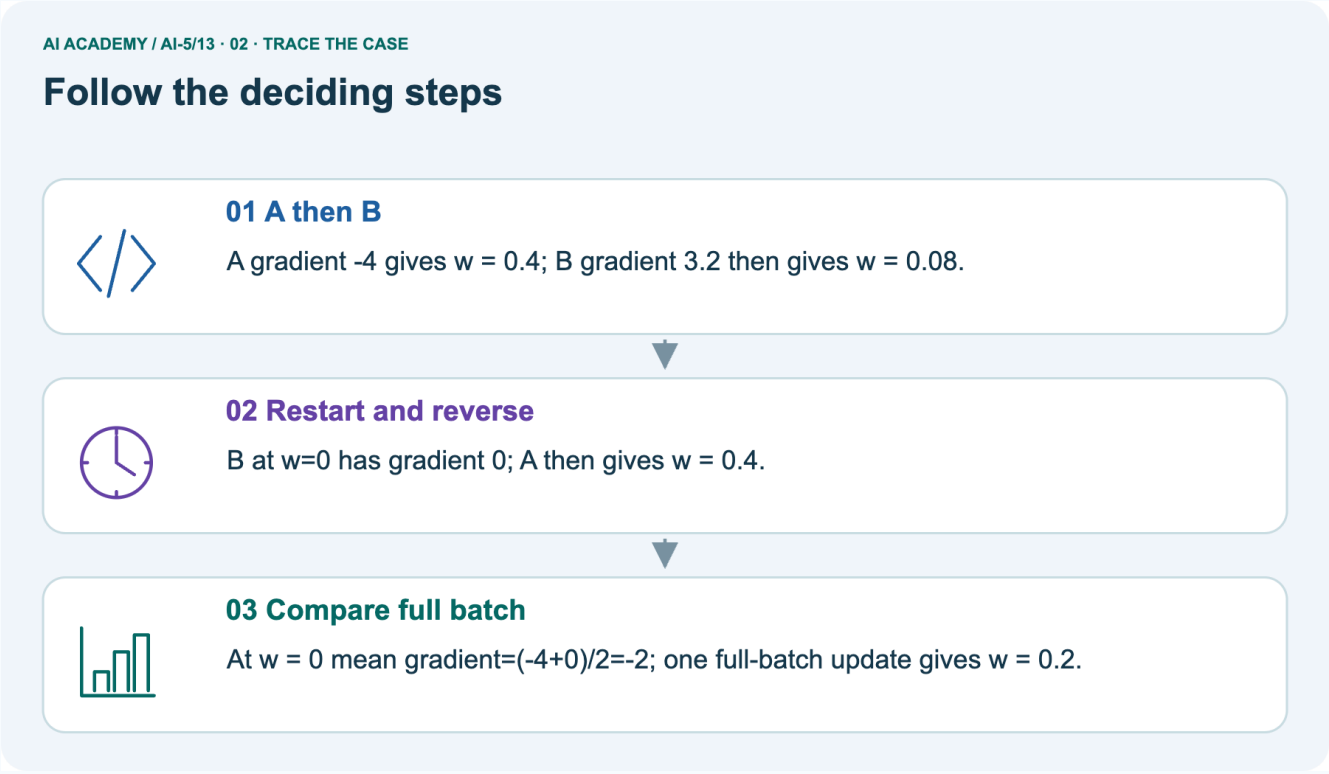
Show why an **epoch**'s example order can matter for single-example **update**s. Supplied facts: Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

A then B: A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08. → Restart and reverse: B at w=0 has gradient 0; A then gives w = 0.4. → Compare full **batch**: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One pass can contain different numbers of updates

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One pass can contain different numbers of updates

Batch rule	Parameter after each batch	Updates
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

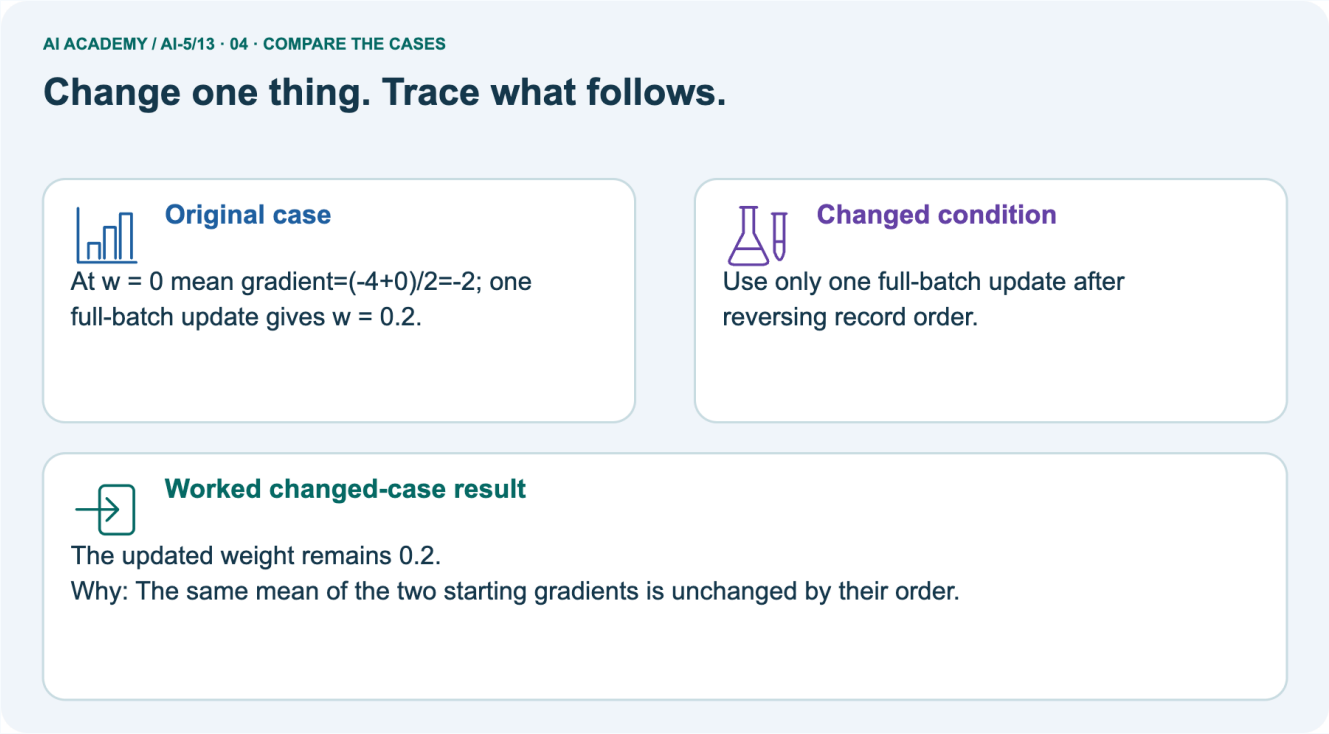
Follow the labelled picture with the matching explanation. Open the vector diagram

One pass can contain different numbers of **update**s. A then B separately | 0 to 0.4 to 1.68 | 2. A and B together | 0 to 1 | 1. B then A separately | 0 to 1.6 to 1.68 | 2

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2. Changed condition: Use only one full-**batch update** after reversing record order. Expected result: The **update**d weight remains 0.2.

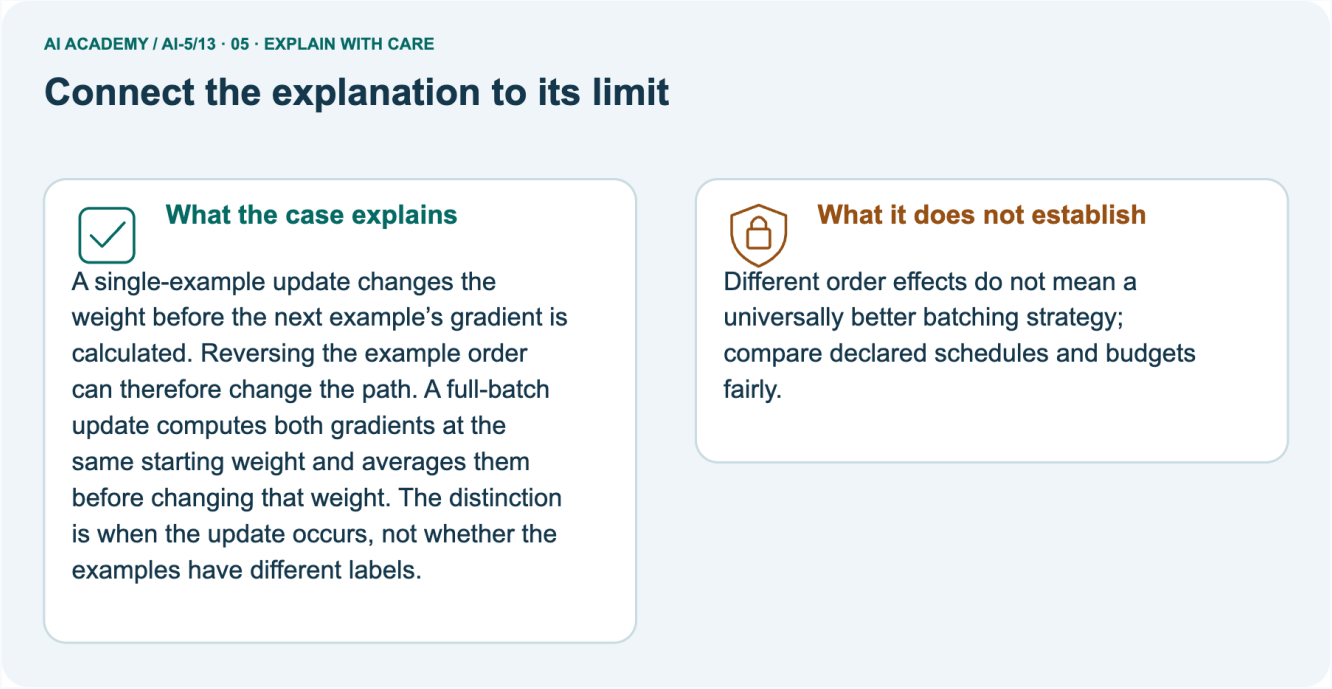
Reason: The same mean of the two starting gradients is unchanged by their order.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The same mean of the two starting gradients is unchanged by their order.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

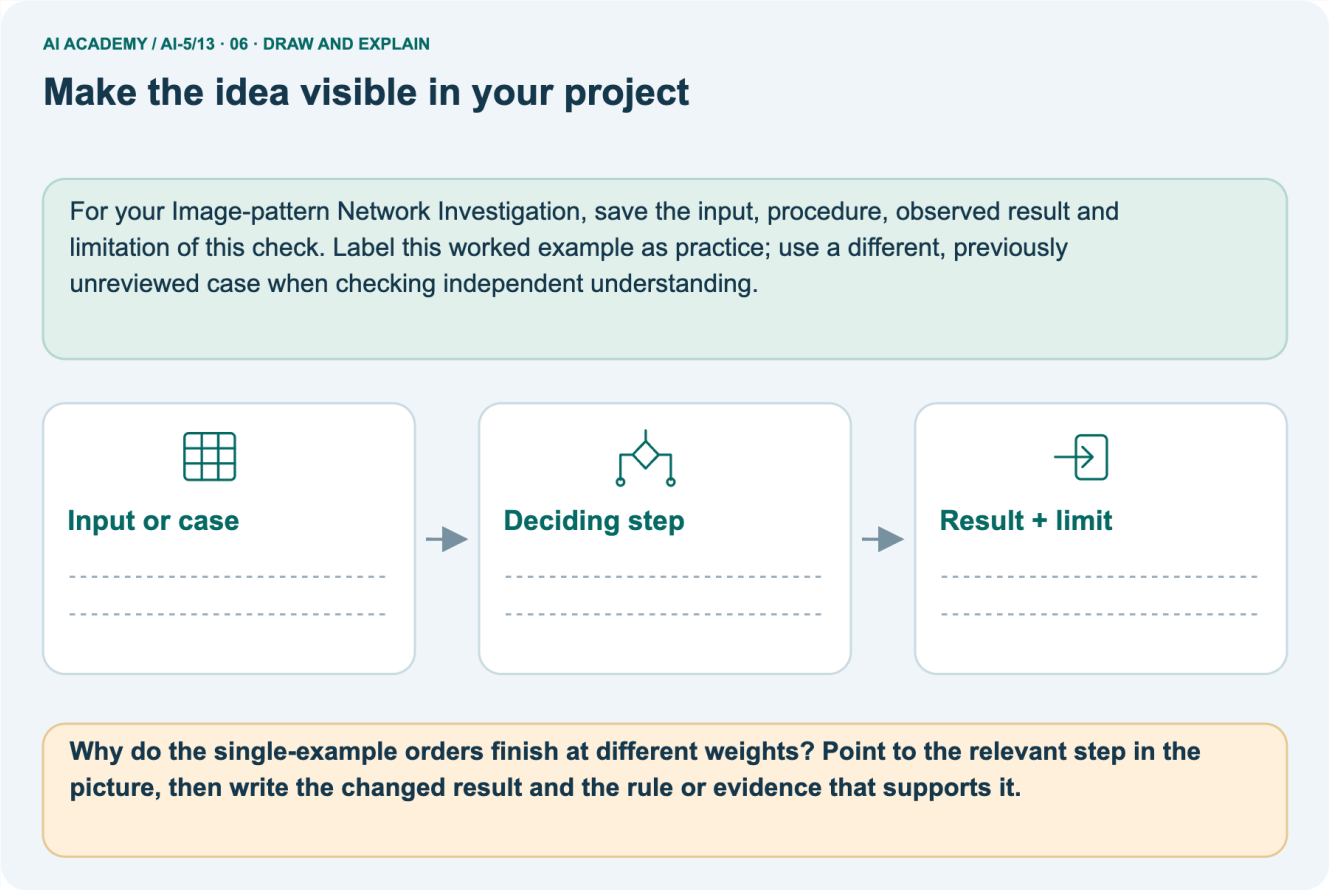
Limit: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The second gradient is evaluated at an already **update**d weight. Full-**batch** gradients here share the same starting weight before averaging. Changed-case check: The **update**d weight remains 0.2.

The same mean of the two starting gradients is unchanged by their order.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

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02. Follow the deciding steps

03. One pass can contain different numbers of updates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.